

UTM TENSILE REPORT — UTM_Test_20260824_104255

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

25.9 MPa

UTS

15.9 MPa

Yield σ_y

1.78 GPa

Modulus E

8.0 %

Failure ϵ_f

1416 kJ/m³

Toughness

-1044 N

Preload anchor

TEST SETTINGS

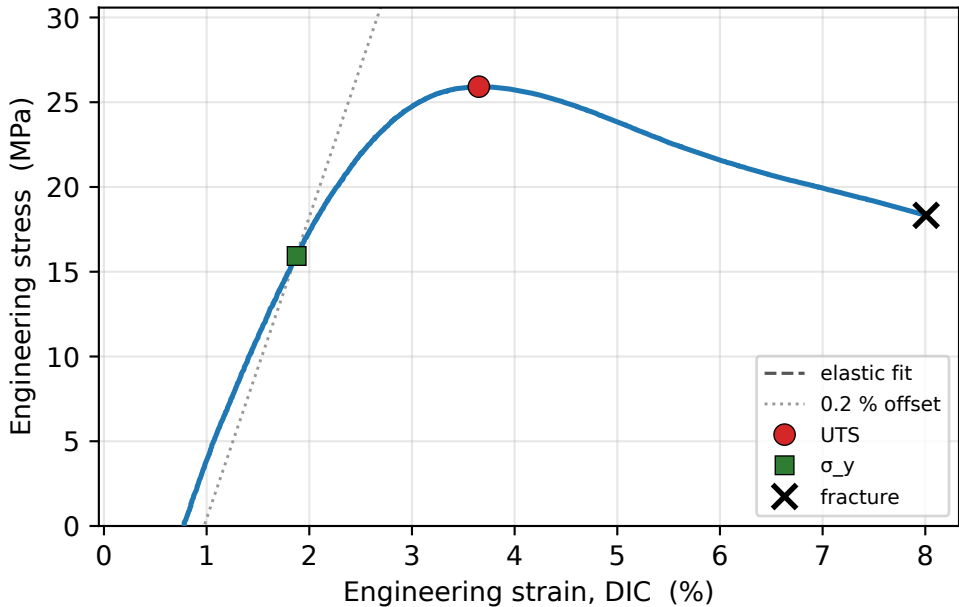
Specimen	UTM_Test_20260824_104255
DIC mode	Black
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 337.616
Comment	—
Test date	2026-08-24 10:40:17
Duration	170.2 s

VALIDATION vs REFERENCES

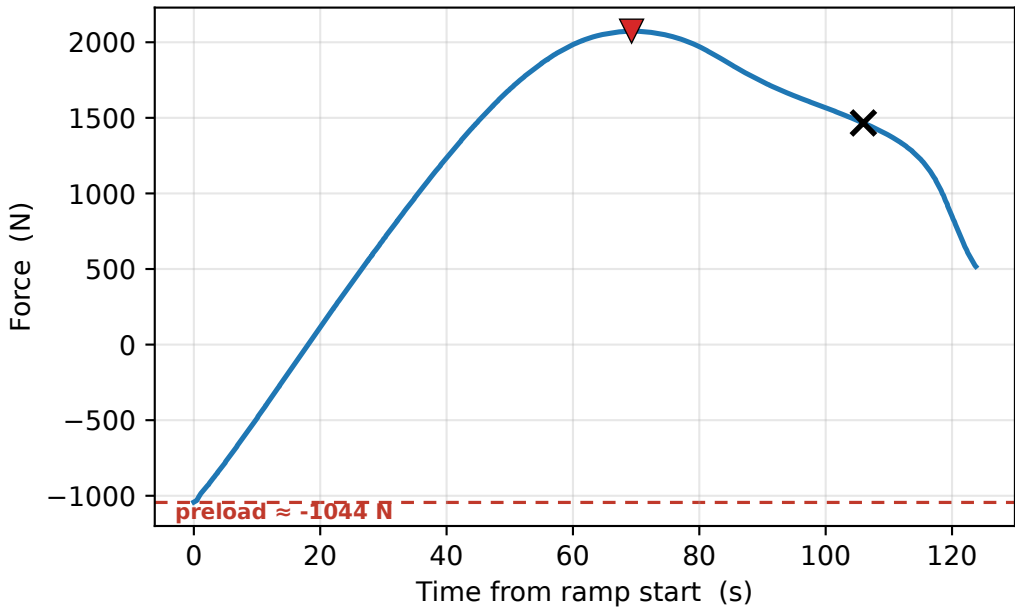
prop	meas	Chacón	E-PLA k
UTS	25.9	32-60 low	×2.24
σ_y	15.9	30-50 low	×3.64
E	1.8	3-5.5 low	×1.61
ϵ_f	8.0 %	(range 3-7 %)	×1.00

⚠ strength outside journal range

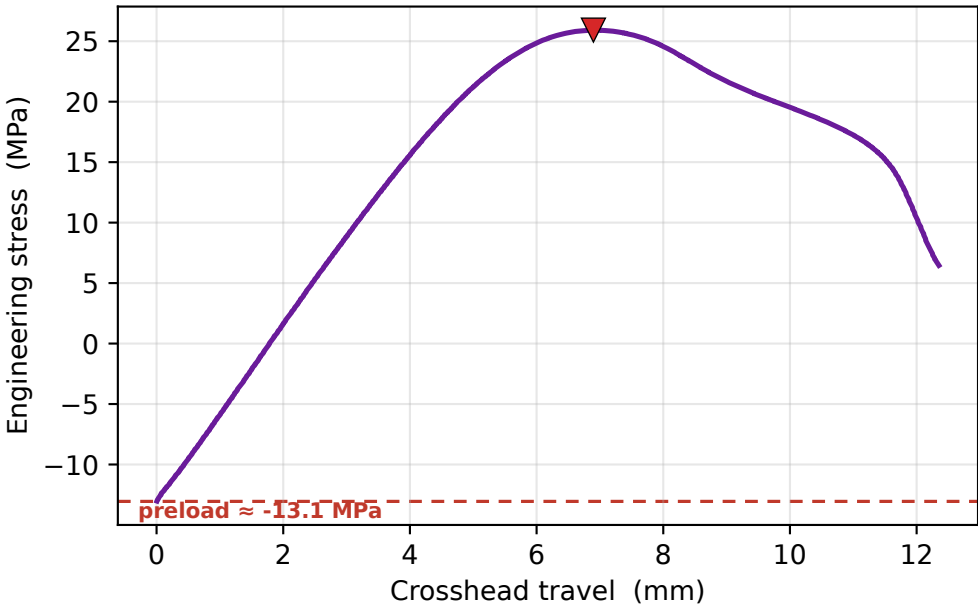
Stress-strain



Load vs time



Stress vs crosshead



DIC strain vs crosshead — the compliance gap

